Chapter 2: One-Dimensional Kinematics

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PHY201

Lecture 2

Part I

Introduction to Quantities Used in Kinematics

Outline of Part I: Introduction to Quantities Used in Kinematics

Introduction to 1-D Kinematics

Kinematics is defined as the study of motion without considering its causes.

κῑ́νημᾰ [kǐː.nɛː.ma] *noun* – Ancient Greek

movement

Descendants: "**kinematics**" and "cinema"

In this chapter, we will study one-dimensional motion of an object without worrying about what forces cause or change its motion.

In the next chapter, we will apply concepts developed here to study two-dimensional motion.

Outline of Part I: Introduction to Quantities Used in Kinematics

Position & Reference Frames

In order to describe the motion of an object, you must first be able to describe its position, where it is at any particular time, relative to a *reference frame*.

Reference Frames

- Dr. Deskins' position could be described in terms of where he is in relation to his desk, which is stationary in Earth's reference frame.
- To describe the position of a person in an airplane, for example, we would use the airplane, not the Earth, as the reference frame.

Displacement ($\Delta\vec{x}$) — Change in Position

If an object moves relative to a reference frame, then the object's position changes.

This change in position is known as **displacement**.

The word "displacement" implies that an object has been displaced (i.e. moved).

Displacement ($\Delta\vec{x}$) — SI Unit: meter (m)

Displacement is the *change in position* of an object:

$$\Delta\vec{x} = \vec{x}_f - \vec{x}_0$$

where $\Delta\vec{x}$ is displacement, $\vec{x}_f$ is the final position, and $\vec{x}_0$ (or $\vec{x}_i$) is the initial position.

Note: Δ always means "change in" whatever quantity follows it.

Note: $\vec{x}_0$ and $\vec{x}_i$ are equivalent.

Example: Displacement

Jeff the monkey moves a crate of bananas from the 2-m mark to the 4-m mark.

What is the crate's displacement?

Recognize from the problem statement:

$$\vec{x}_f = 4\,\mathrm{m}$$

$$\vec{x}_0 = 2\,\mathrm{m}$$

$$\Delta\vec{x} = \vec{x}_f - \vec{x}_0$$

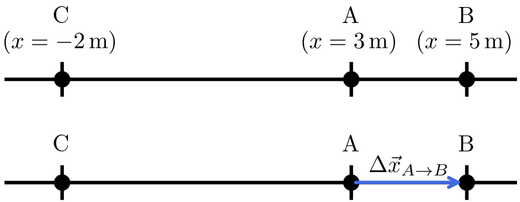
$$\Delta\vec{x} = 4\,\mathrm{m} - 2\,\mathrm{m}$$

$$\boxed{\Delta\vec{x} = 2\,\mathrm{m}}$$

Example: Displacement v. Distance

Heracles the hippo moves from A to B to C.

Calculate (a) his displacement from A to B, (b) his displacement from B to C, (c) his net displacement from A to C, and (d) the total distance he travels.



(a) $\Delta\vec{x}_{A\to B} = \vec{x}_B - \vec{x}_A$

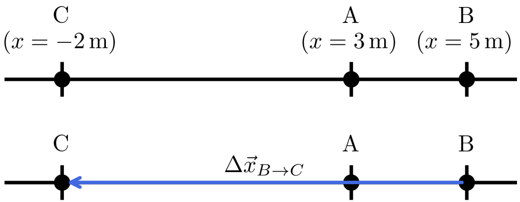
$$\Delta\vec{x}_{A\to B} = 5\,\mathrm{m} - 3\,\mathrm{m}$$

$$\boxed{\Delta\vec{x}_{A\to B} = 2\,\mathrm{m}}$$

Example: Displacement v. Distance

Heracles the hippo moves from A to B to C.

Calculate (a) his displacement from A to B, (b) his displacement from B to C, (c) his net displacement from A to C, and (d) the total distance he travels.



(b) $$\Delta\vec{x}_{B\to C} = \vec{x}_C - \vec{x}_B$$

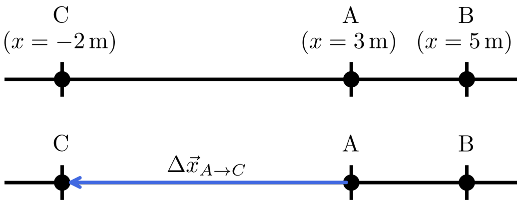
$$\Delta\vec{x}_{B\to C} = -2\,\mathrm{m} - 5\,\mathrm{m}$$

$$\boxed{\Delta\vec{x}_{B\to C} = -7\,\mathrm{m}}$$

Example: Displacement v. Distance

Heracles the hippo moves from A to B to C.

Calculate (a) his displacement from A to B, (b) his displacement from B to C, (c) his net displacement from A to C, and (d) the total distance he travels.

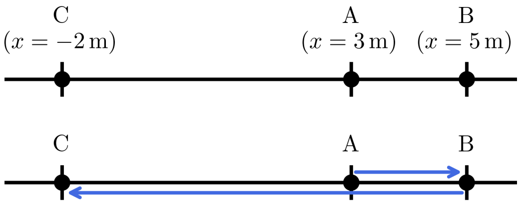


(c)
$$\Delta\vec{x}_{A\to C} = \vec{x}_C - \vec{x}_A$$
$$\Delta\vec{x}_{A\to C} = -2\,\mathrm{m} - 3\,\mathrm{m}$$
$$\boxed{\Delta\vec{x}_{A\to C} = -5\,\mathrm{m}}$$
$$\Delta\vec{x}_{A\to C} = \Delta\vec{x}_{A\to B} + \Delta\vec{x}_{B\to C}$$
$$\Delta\vec{x}_{A\to C} = 2\,\mathrm{m} + (-7\,\mathrm{m}) = -5\,\mathrm{m}$$

Example: Displacement v. Distance

Heracles the hippo moves from A to B to C.

Calculate (a) his displacement from A to B, (b) his displacement from B to C, (c) his net displacement from A to C, and (d) the total distance he travels.



(d) $d = |\Delta\vec{x}_{A\to B}| + |\Delta\vec{x}_{B\to C}|$

$$d = |2\,\mathrm{m}| + |-7\,\mathrm{m}|$$

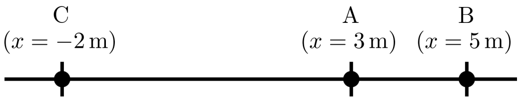
$$d = 2\,\mathrm{m} + 7\,\mathrm{m}$$

$$\boxed{d = 9\,\mathrm{m}}$$

Example: Displacement v. Distance

Heracles the hippo moves from A to B to C.

Calculate (a) his displacement from A to B, (b) his displacement from B to C, (c) his net displacement from A to C, and (d) the total distance he travels.



Notice that the magnitude of Heracles' net displacement is not the same as his distance traveled.

$$\text{if direction is constant, } |\Delta\vec{x}_{net}| = d$$

$$\text{but} \because \text{direction changed, } |\Delta\vec{x}_{net}| \neq d$$

Check Your Understanding

A cyclist rides 3 km west and then turns around and rides 2 km east.

(a) What is their displacement?

(b) What distance do they ride?

(c) What is the magnitude of their displacement?

Come to lecture to see the answer.

Outline of Part I: Introduction to Quantities Used in Kinematics

Vectors v. Scalars

Definitions of "Vector" & "Scalar"

- **Vector:** Quantity with magnitude and direction
- **Scalar:** Quantity with magnitude only (no direction)

Examples of Vectors & Scalars

- Vector quantities: displacement, velocity, force
- Scalar quantities: distance, temperature, speed, energy

Displacement v. Distance

- Displacement = change in position (vector)
- Distance = total distance traveled along a path (scalar)

Vector Etymology

***weǵʰ** *Root* – Proto-Indo-European

to bring; to transport

Descendants:

- *Proto-Germanic* — **wegaz**: way, path
- *Proto-Germanic* — **wagnaz**: cart, wagon
- *Latin* — **vehiculum**: means of transport, vehicle
- *Latin* — **vector**: bearer, carrier

Check your Understanding

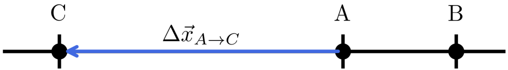
A person's speed can stay the same as they round a corner and changes direction.

Given this information, is speed a scalar or a vector quantity? Explain.

Come to lecture to see the answer.

Representing Vectors

- Vectors have **(1) magnitude** and **(2) direction**.
- Vectors are represented by arrows.
- Arrow length $\propto$ magnitude of the vector (Note: $\propto$ means "proportional to").
- Arrow orientation shows the direction of the vector.



In this example, $\Delta\vec{x}_{A\to C}$ is the displacement vector.

It is represented by a blue arrow.

The length of the arrow shows the magnitude of the vector and the direction of the arrow ($-x$ direction) shows the direction of $\Delta\vec{x}_{A\to C}$.

Representing Vectors

- Vectors have **(1) magnitude** and **(2) direction**.
- Vectors are represented by arrows.
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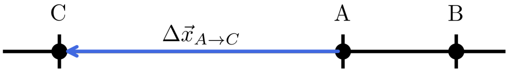


In 1-D, vectors pointing in the positive direction are positive. Vectors in the negative direction are negative.

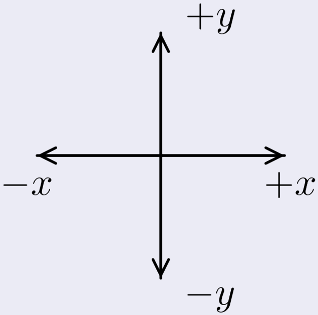
In this example, $\boxed{\Delta\vec{x}_{A\to C} = -5\,\mathrm{m}}$

Since $\Delta\vec{x}_{A\to C}$ is negative, the blue arrow must point in the negative x direction.

Coordinate Systems

When solving a problem, you need to define your coordinate system.

Conventional Coordinate System



- The conventional coordinate system is most common.
- You may choose to define and use a different coordinate system.
- Different coordinate systems may be useful for certain problems.

Be consistent!

You must not switch coordinate systems mid-problem.

Outline of Part I: Introduction to Quantities Used in Kinematics

Introduction

Displacement

Vectors & Scalars

Time, Velocity, and Speed

Acceleration

"What is time? Who can readily and briefly explain this? Who can even in thought comprehend it, so as to utter a word about it? But what in discourse do we mention more familiarly and knowingly, than time? And, we understand, when we speak of it; we understand also, when we hear it spoken of by another.
What then is time? If no one asks me, I know: if I wish to explain it to one that asketh, I know not."

— St. Augustine of Hippo
Confessions, Book XI ($\sim$ 397 A.D.)

Time

Time (t) — SI Unit: second (s)

Time is the interval over which **change** occurs.

Elapsed time Δt is the difference between the ending time t_f and beginning time t_0,

$$\Delta t = t_f - t_0$$

Every measurement of time involves measuring a change in some quantity.

E.g. a number on a digital clock, a heartbeat, or the position of the Sun.

Time

Time (t) — SI Unit: second (s)

Time is the interval over which **change** occurs.

Elapsed time Δt is the difference between the ending time t_f and beginning time t_0,

$$\Delta t = t_f - t_0$$

(Δ means the change in the quantity that follows it.)

For the sake of simplicity, we often say $t_0 = 0$ (just like a stopwatch begins at 0), and if $t_0 = 0$, then $\Delta t = t_f = t$.

Velocity

The **velocity** of an object describes both its speed and its direction of motion.

vēlōx [ˈweː.ɫoːks] *adjective* – Latin

swift, quick

vēlōx is especially used for running, swimming, flying (moving in a **direction**)

Velocity

Average Velocity $(\vec{v}_{\text{avg}})$ — SI Unit: meter per second $\left(\frac{\text{m}}{\text{s}}\right)$

Average velocity is displacement (change in position) divided by the time of travel,

$$\vec{v}_{\text{avg}} = \overline{\boldsymbol{v}} = \frac{\Delta\vec{x}}{\Delta t} = \frac{\vec{x}_f - \vec{x}_0}{t_f - t_0}$$

where $\vec{v}_{\text{avg}}$ or $\overline{\boldsymbol{v}}$ is the average (indicated by the bar over the v) velocity, $\Delta\vec{x}$ is the change in position (or displacement), and x_f and x_0 are the final and initial positions at times t_f and t_0, respectively.

If we choose to make the starting time $t_0 = 0$, then the average velocity is simply

$$\vec{v}_{\text{avg}} = \overline{\boldsymbol{v}} = \frac{\Delta\vec{x}}{t}$$

Velocity

Notice that this definition indicates that **velocity is a vector because displacement is a vector**. It has both magnitude and direction.

$$\vec{v}_{\text{avg}} = \frac{\Delta\vec{x}}{t}$$

Example: Calculating Average Velocity

Calculating Average Velocity

A ball rolls from an initial position $\vec{x}_0 = 2\,\mathrm{m}$ to a final position $\vec{x}_f = 10\,\mathrm{m}$ in a time interval of $\Delta t = 4\,\mathrm{s}$. Find its average velocity.

$$\vec{v}_{\text{avg}} = \frac{\Delta \vec{x}}{\Delta t} = \frac{\vec{x}_f - \vec{x}_0}{t}$$

$$\vec{v}_{\text{avg}} = \frac{10\,\mathrm{m} - 2\,\mathrm{m}}{4\,\mathrm{s}} = \frac{8\,\mathrm{m}}{4\,\mathrm{s}}$$

$$\boxed{\vec{v}_{\text{avg}} = 2\,\frac{\mathrm{m}}{\mathrm{s}}}$$

The positive sign of the velocity means that ball moved in the $+x$ direction.

- What is the ball's displacement?
- Is displacement a vector or scalar?

Velocity v. Speed

Velocity	Vector	Has both magnitude and direction	$\frac{\mathrm{m}}{\mathrm{s}}$
Speed	Scalar	Has magnitude only (no direction)	$\frac{\mathrm{m}}{\mathrm{s}}$

- Magnitude: both quantities answer the question "how fast?"
- Direction: only velocity answers the question "where is it going?"

Example: Velocity vs. Speed

Velocity vs. Speed

A dog runs 8 m to the right in 4 s, then 8 m to the left in 4 s.
Calculate (a) its average velocity and (b) its average speed.

$$\vec{v}_{\text{avg}} = \frac{\Delta\vec{x}}{\Delta t} = \frac{x_f - x_0}{\Delta t}$$

$$\Delta\vec{x} = x_f - x_0 = x_0 - x_0 = 0\,\text{m}$$

$$\vec{v}_{\text{avg}} = \frac{0\,\text{m}}{8\,\text{s}}$$

$$\boxed{\vec{v}_{\text{avg}} = 0\,\frac{\text{m}}{\text{s}}}$$

Velocity depends on **displacement** (includes direction), while speed depends on **distance** (no direction).

Example: Velocity vs. Speed

Velocity vs. Speed

A dog runs 8 m to the right in 4 s, then 8 m to the left in 4 s. Calculate (a) its average velocity and (b) its average speed.

$$\Delta\vec{x}_{\text{net}} = \Delta\vec{x}_1 + \Delta\vec{x}_2$$

$$\Delta\vec{x}_{\text{net}} = 8\,\text{m} + (-8\,\text{m}) = 0\,\text{m}$$

$$\vec{v}_{\text{avg}} = \frac{\Delta\vec{x}_{\text{net}}}{\Delta t} = \frac{0\,\text{m}}{8\,\text{s}}$$

$$\boxed{\vec{v}_{\text{avg}} = 0\,\frac{\text{m}}{\text{s}}}$$

Velocity depends on **displacement** (includes direction), while speed depends on **distance** (no direction).

Example: Velocity vs. Speed

Velocity vs. Speed

A dog runs 8 m to the right in 4 s, then 8 m to the left in 4 s. Calculate (a) its average velocity and (b) its average speed.

Distance: $d = 8\,\mathrm{m} + 8\,\mathrm{m} = 16\,\mathrm{m}$

$$\mathrm{Speed}_{\mathrm{avg}} = \frac{d}{\Delta t} = \frac{16\,\mathrm{m}}{8\,\mathrm{s}}$$

$$\boxed{\mathrm{Speed}_{\mathrm{avg}} = 2\,\frac{\mathrm{m}}{\mathrm{s}}}$$

Velocity depends on **displacement** (includes direction), while speed depends on **distance** (no direction).

Average Velocity v. Instantaneous Velocity

Average Velocity ($\vec{v}_{\text{avg}}$)

Average velocity is the change in position (displacement) over the total time interval.

It gives a sense of how quickly an object moves from A to B over time.

$$\vec{v}_{\text{avg}} = \frac{\Delta \vec{x}}{\Delta t}$$

- Describes **overall motion over an entire time interval**

Instantaneous Velocity ($\vec{v}$)

Instantaneous velocity is the velocity of an object at a specific moment in time.

It becomes equal to average velocity at infinitely small time spans.

$$\vec{v} = \lim_{\Delta t \to 0} \vec{v}_{\text{avg}} = \lim_{\Delta t \to 0} \frac{\Delta \vec{x}}{\Delta t}$$

- Describes **motion at a specific point in time**

Example: Average Velocity v. Instantaneous Velocity

Average Velocity v. Instantaneous Velocity

The position of a particle is given by $x(t) = 4t^2$ (with x in meters and t in seconds).
(a) Find the average velocity between $t = 1$ s and $t = 3$ s.
(b) Find the instantaneous velocity at $t = 2$ s.

(a) $$\vec{v}_{\text{avg}} = \frac{\Delta\vec{x}}{\Delta t}$$

$$\vec{v}_{\text{avg}} = \frac{\vec{x}(t=3) - \vec{x}(t=1)}{3\,\text{s} - 1\,\text{s}}$$

$$\vec{v}_{\text{avg}} = \frac{36\,\text{m} - 4\,\text{m}}{2\,\text{s}} = \frac{32\,\text{m}}{2\,\text{s}} = \boxed{16\,\frac{\text{m}}{\text{s}}}$$

Find the initial and final positions.

$$\vec{x}_f = \vec{x}(t = 3\,\text{s}) = 4(3)^2 = 36\,\text{m}$$
$$\vec{x}_0 = \vec{x}(t = 1\,\text{s}) = 4(1)^2 = 4\,\text{m}$$

- Note: $\vec{x}(t)$ is "x as a function of t".
- e.g. $y(x) = 2x$. $y(3) = y(x = 3) = 2 \cdot 3 = 6$
- i.e. we *evaluate* y at a value of $x = 3$

Example: Average Velocity v. Instantaneous Velocity

Average Velocity v. Instantaneous Velocity

The position of a particle is given by $x(t) = 4t^2$ (with x in meters and t in seconds).
(a) Find the average velocity between $t = 1$ s and $t = 3$ s.
(b) Find the instantaneous velocity at $t = 2$ s.

(b)

$$\vec{v} = \lim_{\Delta t \to 0} \vec{v}_{\text{avg}} = \lim_{\Delta t \to 0} \frac{\Delta \vec{x}}{\Delta t}$$

$$\vec{v} \approx \frac{17.64 - 14.44}{2.1 - 1.9}$$

$$\vec{v} \approx \frac{3.20}{0.2} = \boxed{16\ \frac{\text{m}}{\text{s}}}$$

Use times close to $t = 2\,\text{s}$ to approximate ($\vec{v}$ becomes equal to $\vec{v}_{\text{avg}}$ at infinitely small time spans).

$$t = 1.9\,\text{s},\ \ 2.1\,\text{s}$$

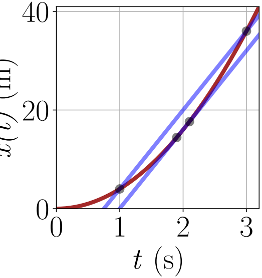
$$\vec{x}_1 = \vec{x}(t = 1.9\,\text{s}) = 4(1.9)^2 = 14.44\,\text{m}$$

$$\vec{x}_2 = \vec{x}(t = 2.1\,\text{s}) = 4(2.1)^2 = 17.64\,\text{m}$$

Example: Average Velocity v. Instantaneous Velocity

Average Velocity v. Instantaneous Velocity

The position of a particle is given by $x(t) = 4t^2$ (with x in meters and t in seconds).
(a) Find the average velocity between $t = 1$ s and $t = 3$ s.
(b) Find the instantaneous velocity at $t = 2$ s.



- For this motion, $\vec{v}$ at the midpoint equals $\vec{v}_{\text{avg}}$ over the interval.
- (This is true for any motion where *acceleration is constant*.)

- What does the slope of the position vs. time graph represent?
- How would the result change for a linear $x(t)$?

Average Velocity v. Average Speed

Average Velocity

Average velocity is the change in position (displacement) over the total time interval.

It is a **vector** quantity, meaning it has both magnitude and direction.

$$\vec{v}_{\text{avg}} = \frac{\Delta \vec{x}}{\Delta t}$$

- Represents the rate at which an object travels from $\vec{x}_0$ to $\vec{x}_f$
- Points from $\vec{x}_0$ to $\vec{x}_f$

Average Speed

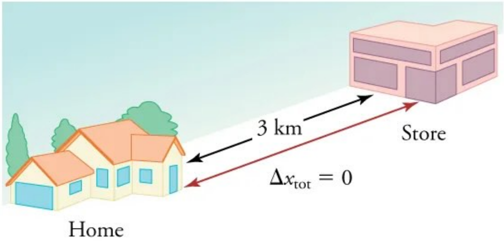
Average speed is the total distance traveled divided by the total time taken.

It is a **scalar** quantity, meaning it only has magnitude and no direction.

$$\text{Speed}_{\text{avg}} = \frac{\text{Distance}}{\Delta t}$$

- Direction does not matter

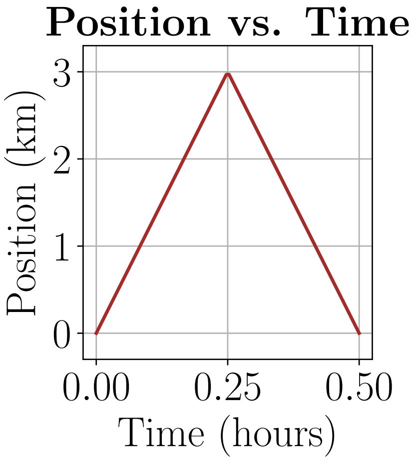
Graphical Depictions of Motion



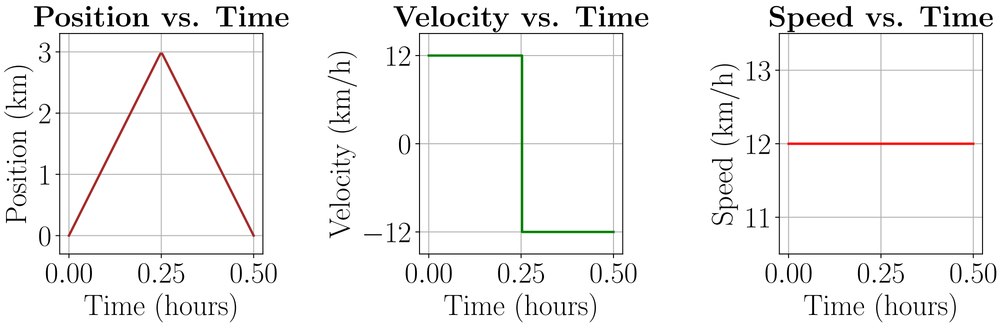
It takes 0.25 h to drive 3 km to the store and 0.25 h to drive back home. Find $\Delta\vec{x}$.

$$\Delta\vec{x} = \vec{x}_f - \vec{x}_0$$

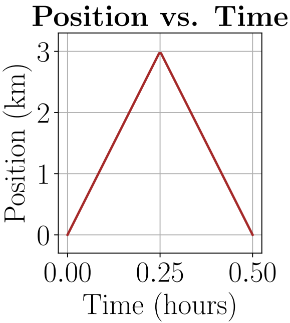
$$\Delta\vec{x} = 0\,\text{km} - 0\,\text{km} = 0$$

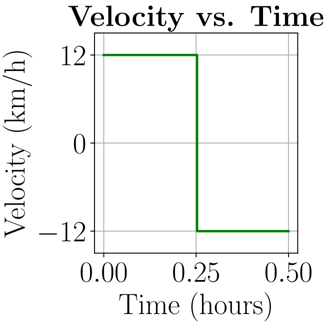


Graphical Depictions of Motion



Graphical Depictions of Motion

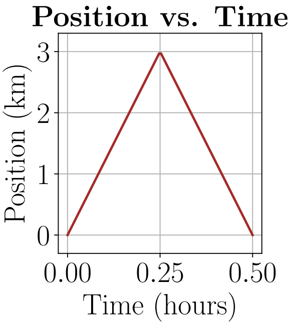


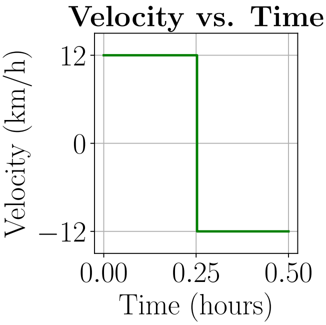


- $\vec{v}$: **slope** of a position v. time plot
- recall $y = mx + b$, a linear equation, where m represents slope

$$m = \frac{\Delta y}{\Delta x} \qquad \vec{v}_{\text{avg}} = \frac{\Delta \vec{x}}{\Delta t}$$

Graphical Depictions of Motion





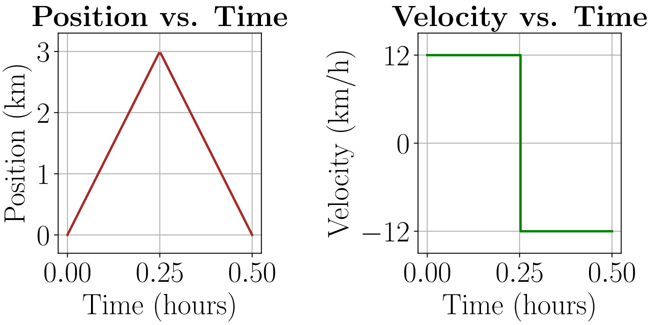
$$\vec{v}_{\text{avg}} = \frac{\Delta \vec{x}}{\Delta t}$$

$$\vec{v}_{\text{avg}} = \frac{3\,\text{km} - 0\,\text{km}}{0.25\,\text{h} - 0\,\text{h}}$$

$$\boxed{\vec{v}_{\text{avg}} = 12\,\frac{\text{km}}{\text{h}}}$$

This is the average velocity between $0\,\text{h}$ and $0.25\,\text{h}$.

Graphical Depictions of Motion



$$\vec{v}_{\text{avg}} = \frac{\Delta \vec{x}}{\Delta t}$$

$$\vec{v}_{\text{avg}} = \frac{0\,\text{km} - 3\,\text{km}}{0.50\,\text{h} - 0.25\,\text{h}}$$

$$\boxed{\vec{v}_{\text{avg}} = -12\,\frac{\text{km}}{\text{h}}}$$

This is the average velocity between 0.25 h and 0.50 h.

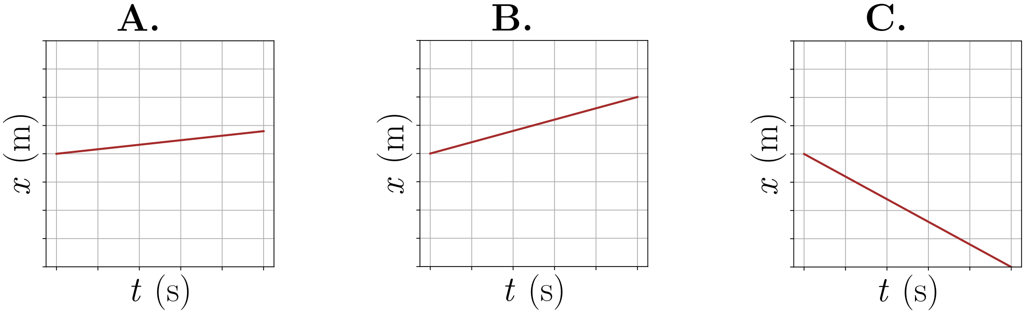
Check Your Understanding

Why was the average velocity $\vec{v}_{\text{avg}}$ negative between $0.25\,\text{h}$ and $0.50\,\text{h}$?

Come to lecture to see the answer.

Check Your Understanding

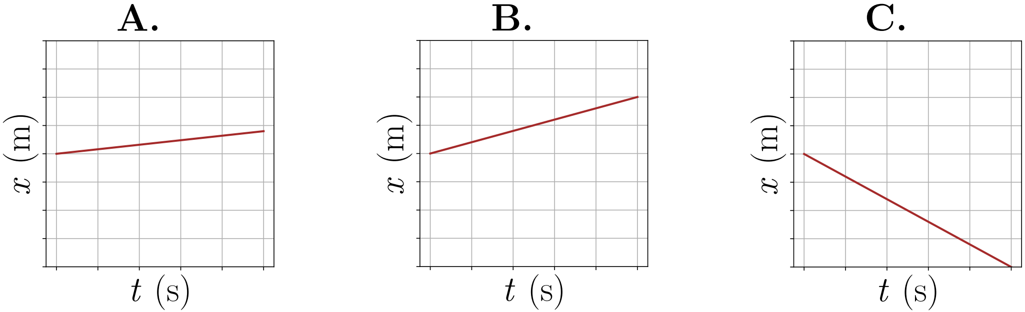
Which of the following objects have a positive velocity?



Come to lecture to see the answer.

Check Your Understanding

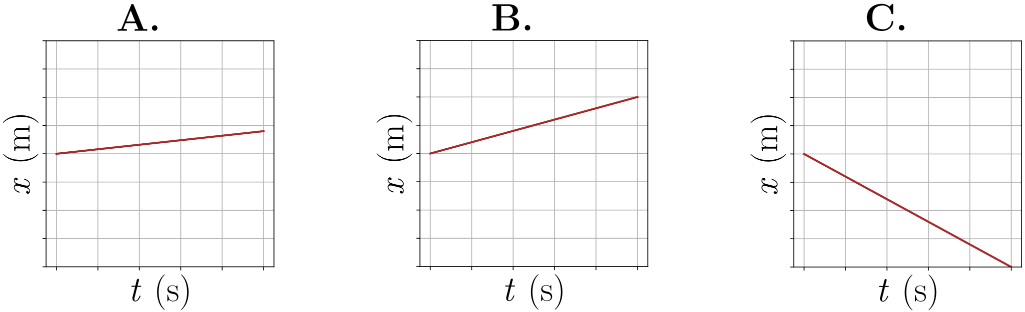
Which of the following objects have a negative velocity?



Come to lecture to see the answer.

Check Your Understanding

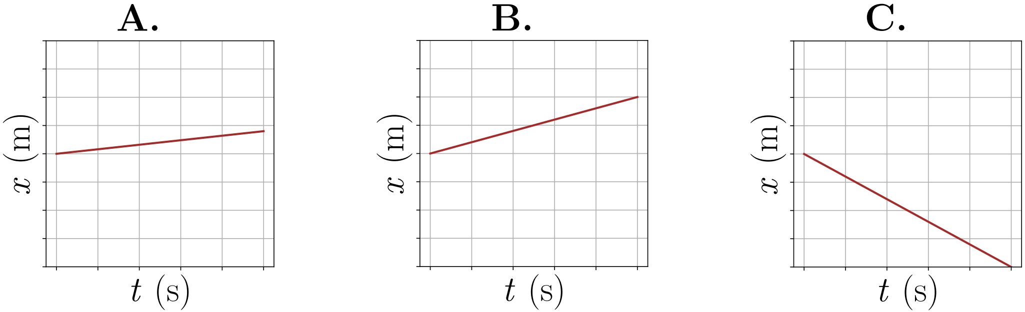
Which of the following objects have a negative speed?



Come to lecture to see the answer.

Check Your Understanding

Which object is traveling fastest?



Come to lecture to see the answer.

Check Your Understanding

A commuter train travels from Baltimore to Washington, DC, and back in 1 hour and 45 minutes. The distance between the two stations is 40 miles.

What is the average velocity of the train?

Come to lecture to see the answer.

Check Your Understanding

A commuter train travels from Baltimore to Washington, DC, and back in 1 hour and 45 minutes. The distance between the two stations is 40 miles.

Calculate the average speed of the train.

Come to lecture to see the answer.

Outline of Part I: Introduction to Quantities Used in Kinematics

Acceleration

Average Acceleration ($\vec{a}_{\text{avg}}$) — SI Unit: (meters per second) per second $\left(\frac{\text{m}}{\text{s}^2}\right)$

Average Acceleration is the rate at which velocity changes,

$$\vec{a}_{\text{avg}} = \overline{\mathbf{a}} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_0}{t_f - t_0}$$

where $\vec{a}_{\text{avg}}$ or $\overline{\boldsymbol{a}}$ is the average (indicated by the bar over the a) acceleration, $\Delta\vec{v}$ is the change in velocity, and v_f and v_0 are the final and initial velocity at times t_f and t_0, respectively.

If we choose to make the starting time $t_0 = 0$, then the average acceleration is simply

$$\vec{a}_{\text{avg}} = \overline{\boldsymbol{a}} = \frac{\Delta\vec{v}}{t}$$

Acceleration

Notice that this definition indicates that **acceleration is a vector because velocity is a vector**. It has both magnitude and direction.

$$\vec{a}_{\text{avg}} = \frac{\Delta \vec{v}}{t}$$

- Acceleration lies in the same direction as the *change in velocity*, $\Delta\vec{v}$.
- Note: acceleration does not always lie in the same direction as *velocity*.
- If acceleration is in the same direction as velocity, we may say the object is accelerating.
- If acceleration is opposite to velocity, we may say the object is decelerating.

Acceleration

- Velocity is a vector.
- Vectors have two parts: (1) magnitude and (2) direction.
- $\therefore$ velocity has two parts: (1) magnitude (speed) and (2) direction of motion.
- Change in something's parts $\iff$ change in the whole thing itself.
- $\therefore$ a change in speed $\implies \Delta\vec{v} \neq 0$
- $\therefore$ a change in direction of motion $\implies \Delta\vec{v} \neq 0$
- Acceleration describes how quickly a change in velocity occurs. $\vec{a}_{\text{avg}} = \Delta\vec{v}/t$
- $\therefore$ a change in speed or direction of motion $\iff \vec{a}_{\text{avg}} \neq 0$
- $\therefore$ constant speed and constant direction $\iff \vec{a}_{\text{avg}} = 0$

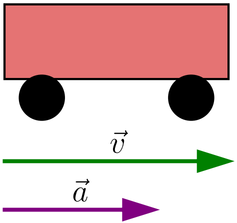
Acceleration and Deceleration

- Deceleration always refers to acceleration in the direction opposite to the direction of the velocity.
- Deceleration always reduces speed.

- Negative acceleration is acceleration in the negative direction in the chosen coordinate system.

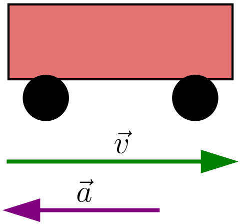
Negative acceleration may or may not be deceleration, and deceleration may or may not be considered negative acceleration.

Acceleration and Deceleration



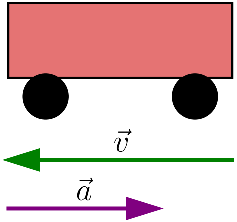
- This car is speeding up as it moves toward the right.
- It therefore has positive acceleration in our coordinate system.

Acceleration and Deceleration



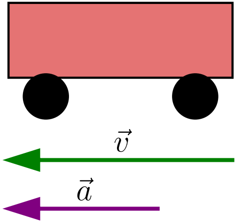
- This car is slowing down as it moves toward the right.
- Therefore, it has negative acceleration in our coordinate system, since its acceleration points left.
- The car is also decelerating: the direction of its acceleration is opposite to its direction of motion.

Acceleration and Deceleration



- This car is moving toward the left, but slowing down over time.
- Therefore, its acceleration is positive in our coordinate system because it is toward the right.
- However, the car is decelerating because its acceleration is opposite to its motion.

Acceleration and Deceleration



- This car is speeding up as it moves toward the left.
- It has negative acceleration because it is accelerating toward the left.
- However, because its acceleration is in the same direction as its motion, it is speeding up (not decelerating).

Check Your Understanding

A ball is thrown upwards.

Just after it is thrown, determine the direction of its:

(a) velocity vector

(b) acceleration vector

Come to lecture to see the answer.

Example: Calculating Acceleration

Calculating Acceleration: A Subway Train Speeding Up — (cp2e_ch2_ex4)

Suppose a train accelerates from rest to $30.0\,\frac{\text{km}}{\text{h}}$ in the first $20.0\,\text{s}$ of its motion. What is its average acceleration during that time interval?

$$\vec{v}_0 = 0,\ \vec{v}_f = 30.0\,\frac{\text{km}}{\text{h}},\ \Delta t = 20.0\,\text{s}$$

Convert all to SI units.

$$\vec{v}_f = 30.0\,\frac{\text{km}}{\text{h}}\left(\frac{1000\,\text{m}}{1\,\text{km}}\right)\left(\frac{1\,\text{h}}{3600\,\text{s}}\right) = 8.33\,\frac{\text{m}}{\text{s}}$$

$$\vec{a}_{\text{avg}} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_0}{t} = \frac{8.33\,\frac{\text{m}}{\text{s}} - 0}{20.0\,\text{s}}$$

$$\boxed{\vec{a}_{\text{avg}} = 0.417\,\frac{\text{m}}{\text{s}^2}}$$

The positive sign indicates acceleration in the positive direction.

- Acceleration has the same direction as $\Delta\vec{v}$.
- The train speeds up from rest, so $\vec{a}$ must be positive.

Example: Calculating Acceleration

Calculating Acceleration: A Subway Train Slowing Down — (cp2e_ch2_ex5)

Now suppose the train slows to a stop from a speed of 30.0 $\frac{\text{km}}{\text{h}}$ in 8.00 s.

What is its average acceleration while stopping?

$$\vec{v}_0 = 30.0\,\frac{\text{km}}{\text{h}},\ \ \vec{v}_f = 0,\ \ \Delta t = 8.00\,\text{s}$$

Convert all to SI units.

$$\vec{v}_0 = 30.0\,\frac{\text{km}}{\text{h}}\left(\frac{1000\,\text{m}}{1\,\text{km}}\right)\left(\frac{1\,\text{h}}{3600\,\text{s}}\right) = 8.33\,\frac{\text{m}}{\text{s}}$$

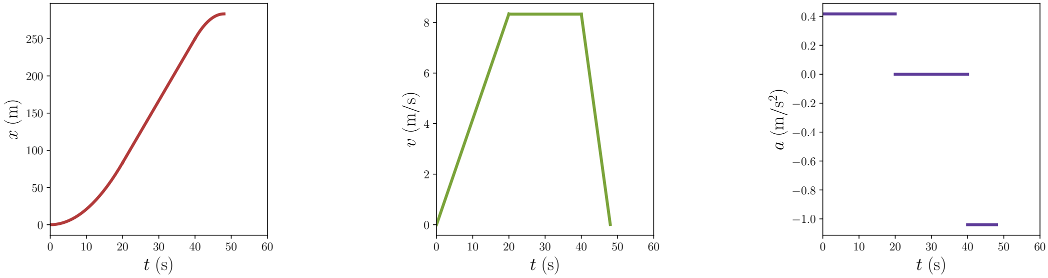
$$\vec{a}_{\text{avg}} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_0}{t} = \frac{0 - 8.33\,\frac{\text{m}}{\text{s}}}{8.00\,\text{s}}$$

$$\boxed{\vec{a}_{\text{avg}} = -1.04\,\frac{\text{m}}{\text{s}^2}}$$

The negative sign indicates acceleration in the direction opposite to the motion.

- Acceleration has the same direction as $\Delta\vec{v}$.
- The train is slowing down, so $\vec{a}$ must be opposite the velocity.

Example: Calculating Acceleration



- slope of a position-versus-time plot at time t is the velocity at time t.
- slope of a velocity-versus-time plot at time t is the acceleration at time t.

Check Your Understanding

An airplane lands on a runway traveling east.
Describe its acceleration.

Come to lecture to see the answer.

Part II

Kinematic Equations

Outline of Part II: Kinematic Equations

Introduction to the Kinematic Equations

Solving Problems using the Kinematic Equations

Equations Used in Kinematics

$$\Delta\vec{x} = \vec{x}_f - \vec{x}_0 \qquad \vec{v}_{\text{avg}} = \frac{\Delta\vec{x}}{\Delta t} \qquad \vec{a}_{\text{avg}} = \frac{\Delta\vec{v}}{\Delta t}$$

$$\vec{v}(t) = \vec{v}_0 + \vec{a}t$$

$$\vec{x}(t) = \vec{x}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}t^2 \qquad h = \frac{v_{0y}^2}{2g}$$

$$\vec{v}(x)^2 = \vec{v}_0^2 + 2\vec{a}\cdot(\vec{x} - \vec{x}_0)$$

$$\vec{v}_{\text{avg}} = \frac{\vec{v}_f + \vec{v}_0}{2} = \frac{\Delta\vec{x}}{\Delta t} \qquad R = \frac{v_0^2 \sin 2\theta_0}{g}$$

Kinematic Equations

$$\vec{v}(t) = \vec{v}_0 + \vec{a}t$$

$$\vec{x}(t) = \vec{x}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}t^2$$

$$v(x)^2 = v_0^2 + 2a(x - x_0)$$

$$\vec{v}_{\text{avg}} = \frac{\vec{v}_f + \vec{v}_0}{2} = \frac{\Delta\vec{x}}{\Delta t}$$

Math Review – Functions

A function assigns an output to each input:

$$y = f(x)$$

This declares that "y is a function of x", which means that y depends on x.

- $y(x)$ means "the value of y at input x"
- Example: $y(x) = 5x^2$
- Evaluation: $y(2) = 5(2^2) = 5(4) = 20$

Outline of Part II: Kinematic Equations

Introduction to the Kinematic Equations

Solving Problems using the Kinematic Equations

Example: Calculating Displacement

Calculating Displacement: How Far Does the Jogger Run? — (cp2e_ch2_ex8)

A jogger runs down a straight stretch of road with an average velocity of $4.00\,\frac{\text{m}}{\text{s}}$ for $2.00\,\text{min}$. What is his final position, taking his initial position to be zero?

$$\vec{x}_0 = 0,\ \vec{v}_{\text{avg}} = 4.00\,\tfrac{\text{m}}{\text{s}},\ \Delta t = 2.00\,\text{min}$$

$$\Delta t = 2.00\,\text{min}\left(\frac{60\,\text{s}}{1\,\text{min}}\right) = 120\,\text{s}$$

$$\vec{v}_{\text{avg}} = \frac{\Delta\vec{x}}{\Delta t}\ \therefore\ \Delta\vec{x} = \vec{v}_{\text{avg}}\Delta t = \vec{x}_f - \vec{x}_0$$

$$\therefore\ \vec{x}_f = \vec{x}_0 + \vec{v}_{\text{avg}}\Delta t = 0 + (4.00\,\tfrac{\text{m}}{\text{s}})(120\,\text{s})$$

$$\boxed{\vec{x}_f = 480\,\text{m}}$$

The positive result indicates the jogger's displacement is in the same direction as the velocity.

Check Your Understanding

A ball falls from a high place.

Following the conventional coordinate system, is the ball's vertical displacement positive or negative?

Come to lecture to see the answer.

Example: Calculating Final Velocity

Calculating Final Velocity: Airplane — (cp2e_ch2_ex9)

An airplane lands with an initial velocity of $70.0\,\frac{\mathrm{m}}{\mathrm{s}}$ and then decelerates at $-1.50\,\frac{\mathrm{m}}{\mathrm{s}^2}$ for $40.0\,\mathrm{s}$. What is its final velocity?

$$\vec{v}_0 = 70.0\,\frac{\mathrm{m}}{\mathrm{s}},\ \ \vec{a} = -1.50\,\frac{\mathrm{m}}{\mathrm{s}^2},\ \ t = 40.0\,\mathrm{s}$$

$$\vec{v}(t) = \vec{v}_0 + \vec{a}t$$

$$\vec{v}(40.0\,\mathrm{s}) = 70.0\,\frac{\mathrm{m}}{\mathrm{s}} + \left(-1.50\,\frac{\mathrm{m}}{\mathrm{s}^2}\right)(40.0\,\mathrm{s})$$

$$\vec{v}_f = \vec{v}(40.0\,\mathrm{s})$$

$$\therefore \quad \boxed{\vec{v}_f = 10.0\,\frac{\mathrm{m}}{\mathrm{s}}}$$

$|\vec{v}_f| < |\vec{v}_0| \ \therefore$ plane moves more slowly. $\vec{v}_f$ is positive, which means the plane is still moving forward.

- Acceleration is opposite velocity $\therefore$ object slows down.
- If $\vec{v}_f$ became negative, then motion would reverse direction.

Check Your Understanding

A ball is thrown straight up into the air.

At the very top of its path, what is its velocity? What is its acceleration?

Come to lecture to see the answer.

Example: Calculating Displacement (Accelerating Object)

Calculating Displacement of an Accelerating Object — (cp2e_ch2_ex10)

Dragsters can achieve average accelerations of $26.0\,\frac{\mathrm{m}}{\mathrm{s}^2}$. Suppose such a dragster accelerates from rest at this rate for $5.56\,\mathrm{s}$. How far does it travel in this time?

$$\vec{x}_0 = 0,\ \vec{v}_0 = 0,\ \vec{a} = 26.0\,\tfrac{\mathrm{m}}{\mathrm{s}^2},\ t = 5.56\,\mathrm{s}$$

$$\vec{x}(t) = \vec{x}_0 + \vec{v}_0\,t + \tfrac{1}{2}\,\vec{a}t^2$$

$$\vec{x}(5.56\,\mathrm{s}) = 0 + 0 + \tfrac{1}{2}(26.0\,\tfrac{\mathrm{m}}{\mathrm{s}^2})(5.56\,\mathrm{s})^2$$

$$\vec{x}_f = \vec{x}(5.56\,\mathrm{s}), \quad \Delta\vec{x} = \vec{x}_f - \vec{x}_0$$

$$\therefore \quad \boxed{\Delta\vec{x} = 402\,\mathrm{m}}$$

The displacement is large for such a short time due to the very high acceleration.

- Starting from rest $\Rightarrow$ $\vec{v}_0 = 0$.
- $402\,\mathrm{m} \approx \frac{1}{4}$ mile (reasonable)

Check Your Understanding

A car moves with constant velocity along a straight road. What is its acceleration?

Come to lecture to see the answer.

Example: Calculating Final Velocity

Calculating Final Velocity: Dragsters — (cp2e_ch2_ex11)

Calculate the final velocity of the dragster in Example 2.10 without using information about time.

$$\vec{v}_0 = 0,\ \vec{a} = 26.0\,\frac{\mathrm{m}}{\mathrm{s}^2},\ \Delta\vec{x} = 402\,\mathrm{m}$$

$$v(x)^2 = v_0^2 + 2a(x - x_0)$$

$$v(402\,\mathrm{m}) = \sqrt{0 + 2(26.0\,\tfrac{\mathrm{m}}{\mathrm{s}^2})(402\,\mathrm{m})}$$

$$v_f = \sqrt{2.09 \times 10^4\,\frac{\mathrm{m}^2}{\mathrm{s}^2}}$$

$$\therefore \quad \boxed{\vec{v}_f = 145\,\frac{\mathrm{m}}{\mathrm{s}}}$$

- Note: $\Delta\vec{x} = \vec{x}_f \because \vec{x}_0 = 0$

- We take the positive square root since the motion is in the same direction as the acceleration.
- Acceleration and displacement in same direction $\Rightarrow \vec{v}_f > 0$.
- $\vec{v}_f \approx 145\,\frac{\mathrm{m}}{\mathrm{s}}$ (over 300 mph!)

Example: Stopping Distance and Reaction Time

Calculating Displacement: Stopping Car — (cp2e_ch2_ex12)

A car moving at $30.0\,\frac{\mathrm{m}}{\mathrm{s}}$ brakes with deceleration $-7.00\,\frac{\mathrm{m}}{\mathrm{s}^2}$ (dry) or $-5.00\,\frac{\mathrm{m}}{\mathrm{s}^2}$ (wet).
(a) Find stopping distances. (b) Include a $0.500\,\mathrm{s}$ reaction time.

$$v_f^2 = v_0^2 + 2a\Delta x$$

$$\Delta\vec{x} = \frac{v^2 - v_0^2}{2a}$$

$$\Delta\vec{x} = \frac{0 - (30.0\,\frac{\mathrm{m}}{\mathrm{s}})^2}{2(-7.00\,\frac{\mathrm{m}}{\mathrm{s}^2})} = 64.3\,\mathrm{m} \quad \text{(Dry)}$$

$$\Delta\vec{x} = \frac{0 - (30.0\,\frac{\mathrm{m}}{\mathrm{s}})^2}{2(-5.00\,\frac{\mathrm{m}}{\mathrm{s}^2})} = 90.0\,\mathrm{m} \quad \text{(Wet)}$$

Reaction distance:

$$\Delta\vec{x} = \vec{v}_{\text{avg}}\,t = (30.0\,\tfrac{\mathrm{m}}{\mathrm{s}})(0.500\,\mathrm{s}) = 15.0\,\mathrm{m}$$

Total distances:

$$\boxed{\Delta\vec{x}_{\text{dry}} = 79.3\,\mathrm{m}}$$

$$\boxed{\Delta\vec{x}_{\text{wet}} = 105\,\mathrm{m}}$$

Example: Calculating Time

Calculating Time: A Car Merges into Traffic — (cp2e_ch2_ex13)

A car travels along a 200 m ramp with initial velocity $10.0\,\frac{\text{m}}{\text{s}}$ and acceleration $2.00\,\frac{\text{m}}{\text{s}^2}$. How long does it take to travel the ramp?

$$\vec{x}_0 = 0,\ \vec{v}_0 = 10.0\,\tfrac{\text{m}}{\text{s}},$$
$$\vec{a} = 2.00\,\tfrac{\text{m}}{\text{s}^2},\ \Delta\vec{x} = 200\,\text{m}$$

$$\vec{x}(t) = \vec{x}_0 + \vec{v}_0 t + \tfrac{1}{2}\vec{a}t^2$$

$$200 = 0 + (10.0)t + \tfrac{1}{2}(2.00)t^2$$

$$200 = 10t + t^2$$

$$0 = t^2 + 10t - 200$$

$$t = \tfrac{-b \pm \sqrt{b^2-4ac}}{2a},\ a = 1,\ b = 10,\ c = -200$$

$$t = \tfrac{-10 \pm \sqrt{10^2-4(1)(-200)}}{2}$$

$$= \tfrac{-10 \pm \sqrt{900}}{2} = \frac{-10 \pm 30}{2}$$

$$t = 10.0\,\text{s},\ -20.0\,\text{s}$$

$$\boxed{t = 10.0\,\text{s}}$$

Notes on Previous Slide's Example

- Remember: quadratic equations yield two mathematical solutions.
- The negative solution is unphysical and is discarded.
- Result (~ 10 s) is reasonable.

Including Units in your Work

- I rarely omit units in my work.
- This example is one in which I omit the units.
- If *many* algebraic steps are anticipated:
  - Ensure all quantities are in SI Units.
  - Omit the units and work carefully.
  - Include units at the end.

Check Your Understanding

You jog to a friend's house 2 miles away and then jog back in thirty minutes, returning to your starting point.

For the round trip, calculate the following:

(a.) average speed

(b.) average velocity

Come to lecture to see the answer.

Check Your Understanding

A car moves in the negative direction and is slowing down.

Is its acceleration positive or negative? Is the car decelerating?

Come to lecture to see the answer.

Check Your Understanding

True or False:

Two cars can have the same speed but different velocities.

Come to lecture to see the answer.

Solving Kinematics Problems

A General Approach

- **What do I know?** — List all given quantities with units.
- **What do I want?** — Identify the unknown you are solving for.
- **Which equation connects them?** — Choose the kinematic equation that relates your knowns to your unknown.

Before You Box Your Answer, Ask:

- Did I define a coordinate system and stick to it?
- Did I convert all quantities to SI units?
- Does the sign of my answer make physical sense?
- Does the magnitude of my answer make physical sense?

Next Lecture: Two-Dimensional Kinematics

Today, we learned to describe motion along a single axis. Every concept has a precise meaning:

- displacement, velocity, and acceleration are **vectors**
- sign encodes direction
- the kinematic equations connect position, velocity, acceleration, and time

Chapter 3: Two-Dimensional Kinematics

- We apply 1-D kinematics independently in the x-direction and the y-direction.
- Time bridges the two directions.
- Projectile motion (e.g. a ball thrown at an angle) is the central application.